

Domain Model

Adding Associations

Chapter 11
Applying UML and Patterns
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Objectives

- Identify associations for a domain model.
- Distinguish between need-to-know and comprehension-only associations.

Introduction

- Identify associations of conceptual classes needed to satisfy the information requirements of current scenarios.
- Also identify the aid in comprehending the domain model.

Associations

An association is a relationship between instances of types that indicates some meaningful and interesting connection

Associations

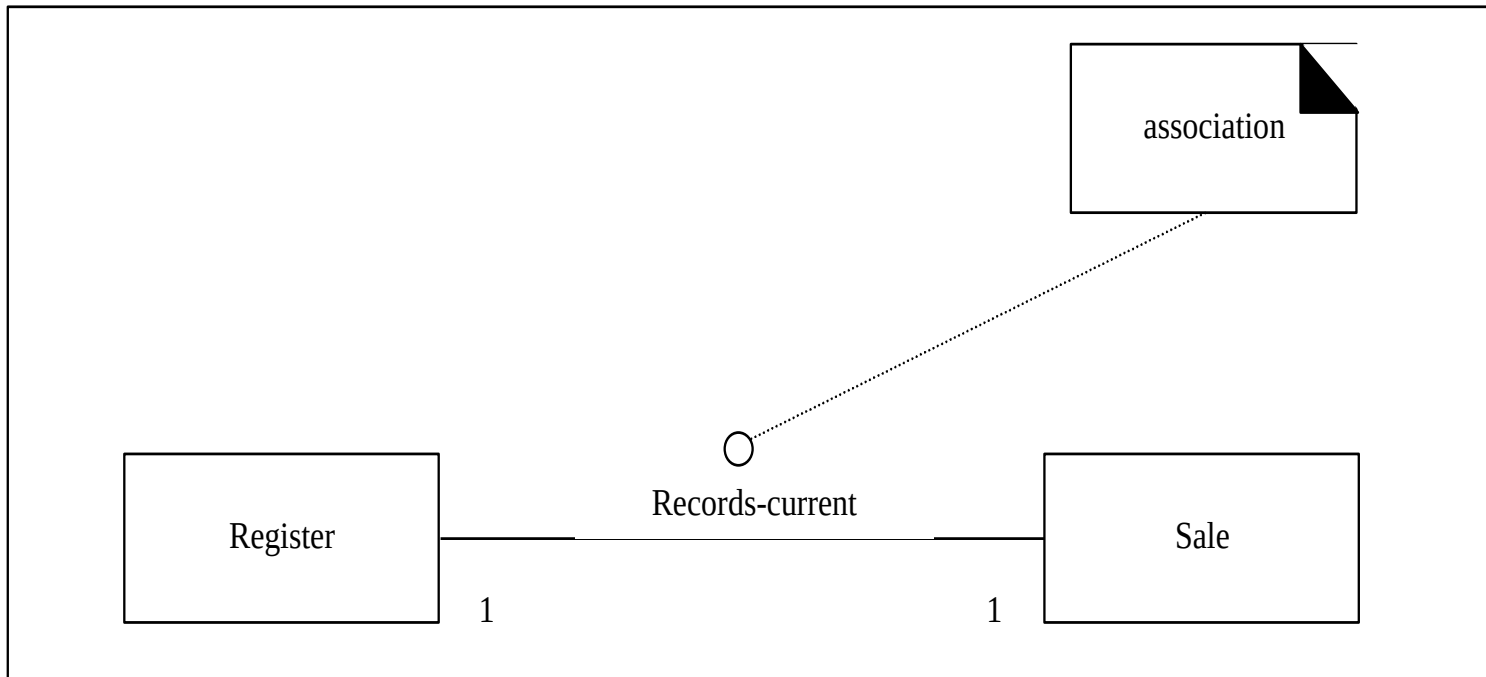


Fig 1.Associations

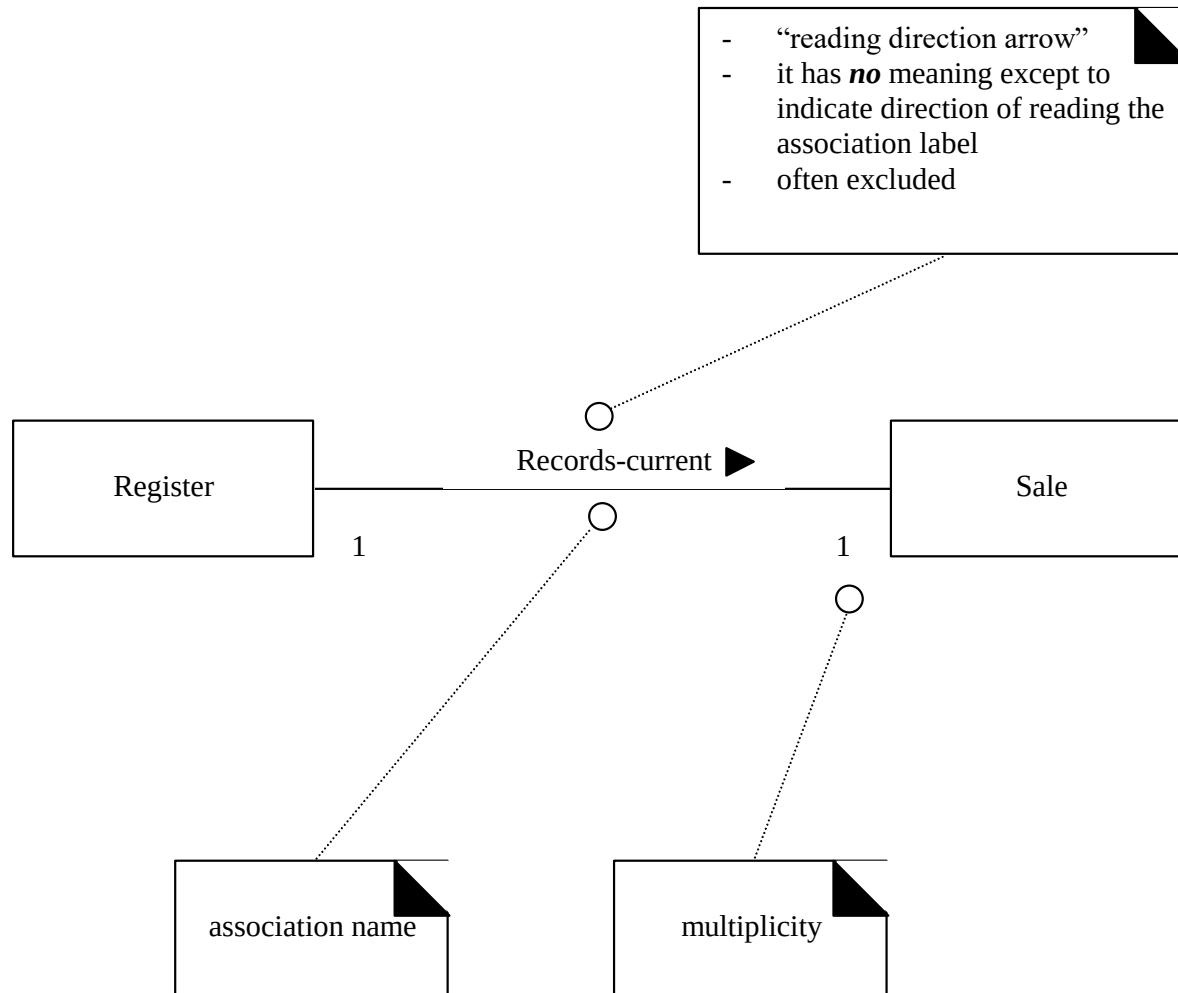
Useful Associations

- Associations for which knowledge of the relationship needs to be preserved for some duration.
- Associations derived from the Common Associations List.

UML Association Notation

- An association is represented as a line between classes with an association name.
- Associations are inherently bidirectional.
- Optional reading direction arrow is only an aid to the reader of the diagram.

UML Association Notation



Finding Associations-Common Associations List

The common categories that are worth considering are:

- A is a physical part of B . *Eg: Wing-Airplane*
- A is a logical part of B. *Eg: SalesLineItem-Sale.*
- A is physically contained in B . *Eg: Register-Store.*

Common Associations List 2

- A is logically contained in B.
Eg:ItemDescription-Catalog.
- A is a description of B.*Eg:ItemDescription-Item.*
- A is a line item of a transaction or report B.*Eg:SalesLineItem-Sale.*
- A is a member of B .*Eg: Cashier-Store.*
- A uses or manages B.*Eg:Cashier-Register.*

Common Associations List 3

- A is known/logged/recorded/reported/captured in B. Eg: Sale-Register.
- A is an organizational subunit of B .
Eg:Department-Store.
- A communicates with B. *Eg:Customer-Cashier.*
- A is next to B. *Eg:City-City.*

Common Associations List 4

- A is related to a transaction B. *Eg: Customer-Payment.*
- A is a transaction related to another transaction B. *Eg: Payment-Sale.*
- A is next to B. *Eg: City-City.*
- A is owned by B. *Eg: Register-Store.*
- A is an event related to B. *Eg: Sale-Customer.*

High-Priority Associations

- A is a physical or logical part of B.
- A is physically or logically contained in/on B.
- A is recorded in B.

Associations Guidelines

- The knowledge of the relationship needs to be preserved for some duration.
- Identifying conceptual classes is more important than identifying associations.
- Avoid showing redundant or derivable associations.

Roles

- Each end of an association is called a role.
- Roles may have
 - *name*
 - *multiplicity expression*
 - *navigability*

Multiplicity

- Multiplicity defines the number of instances of a class A ,that can be associated with one instance of class B,at a particular moment.
- Eg: In countries with monogamy laws,a person can be married to 1 person at any particular time.But over a span of time one person can be married to many persons.

Multiplicity

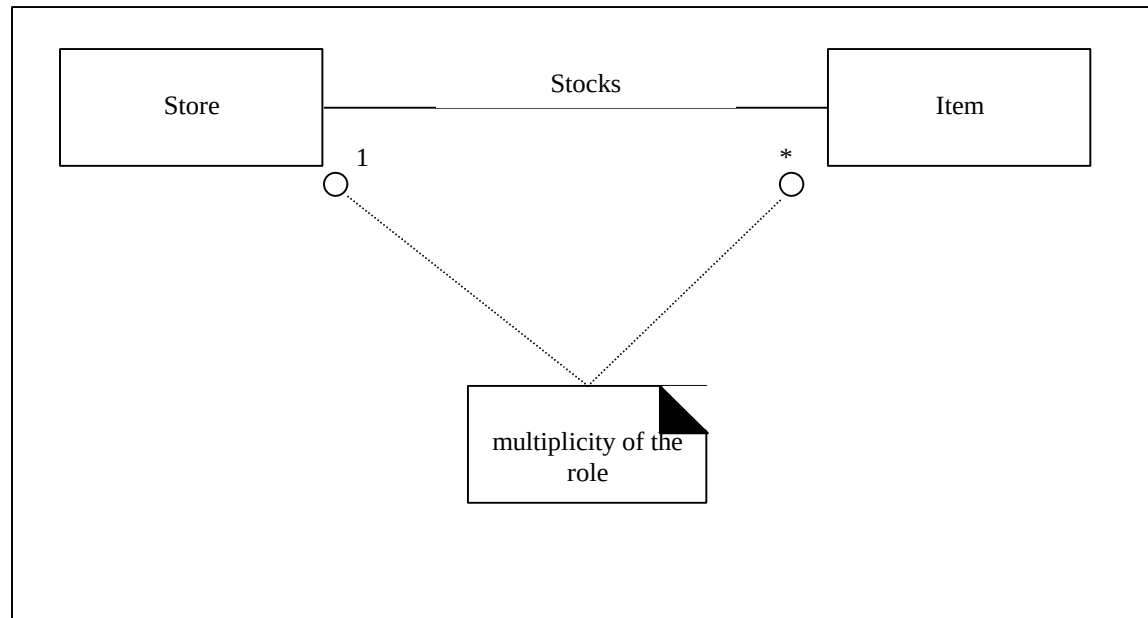


Fig 3 Multiplicity on an association.

Multiplicity

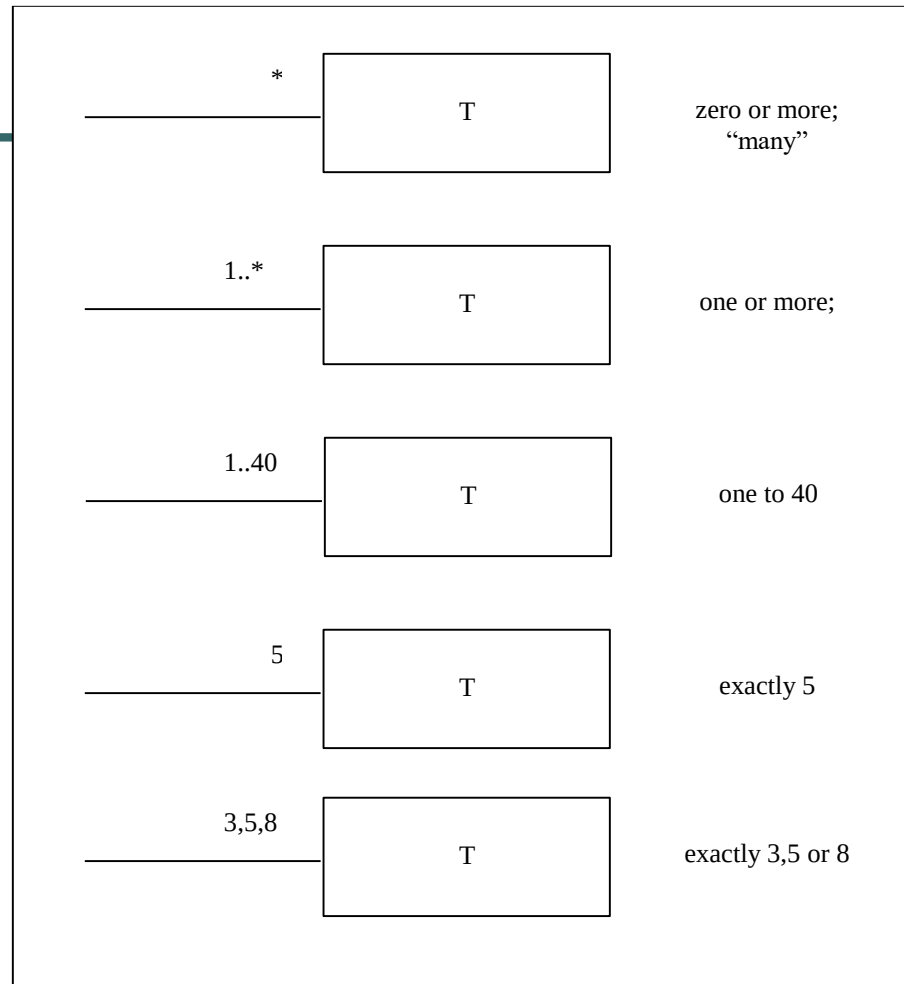


Fig 4. Multiplicity values.

Naming Associations

- Name an association based on TypeName-VerbPhrase-TypeName format.
- Names should start with a capital letter.
- Legal formats are:
 - Paid-by
 - PaidBy

Associations Names

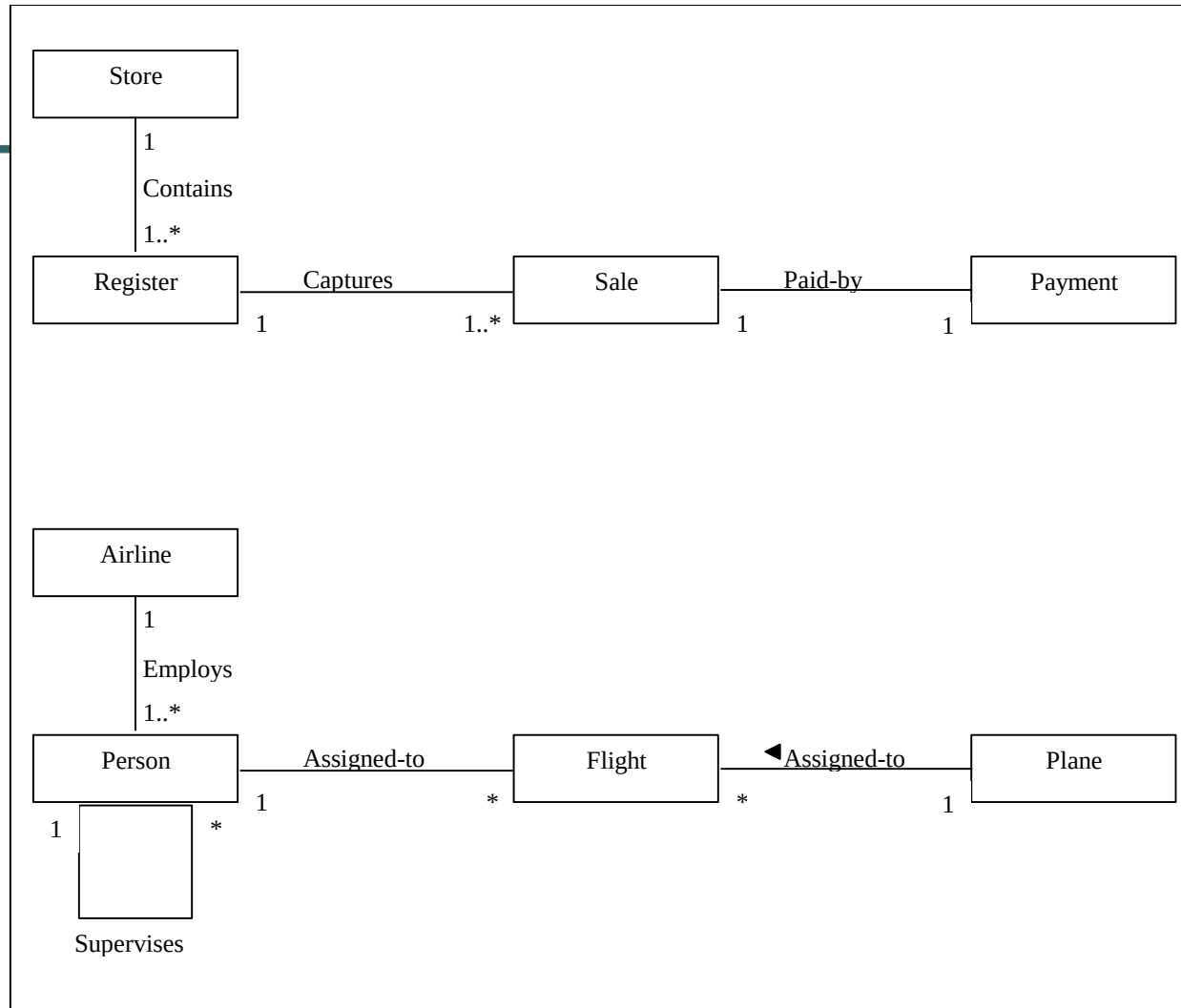


Fig 5.Association names.

Multiple Associations

- Two types may have multiple associations between them.

Multiple Associations

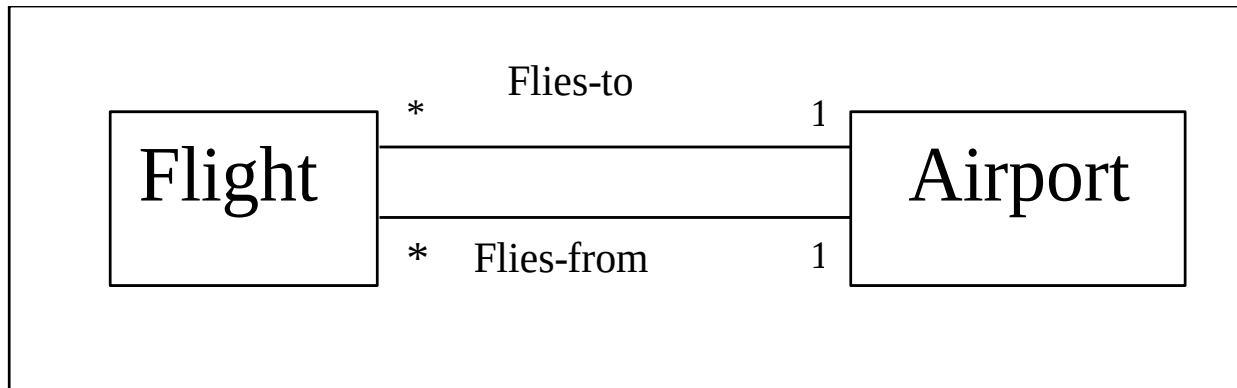


Fig 6. Multiple associations.

Implementation

- The domain model can be updated to reflect the newly discovered associations.
- But, avoid updating any documentation or model unless there is a concrete justification for future use.
- Defer design considerations so that extraneous information is not included and maximizing design options later on.

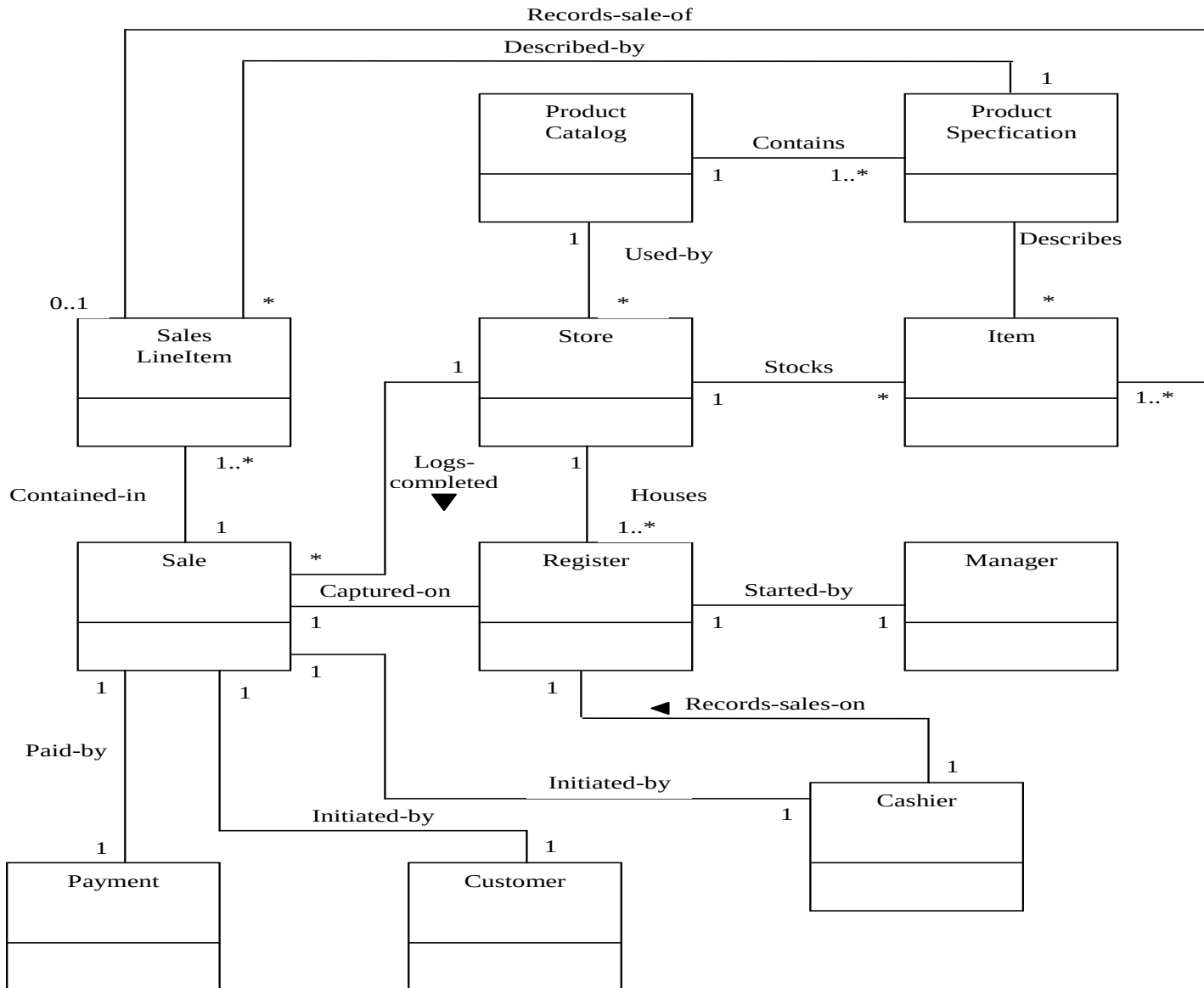
Cleaning Associations 1

- Do not overwhelm the domain model with associations that are not strongly required.
- Use need-to-know criterion for maintaining associations.
- Deleting associations that are not strictly demanded on a need-to-know basis can create a model that misses the point.

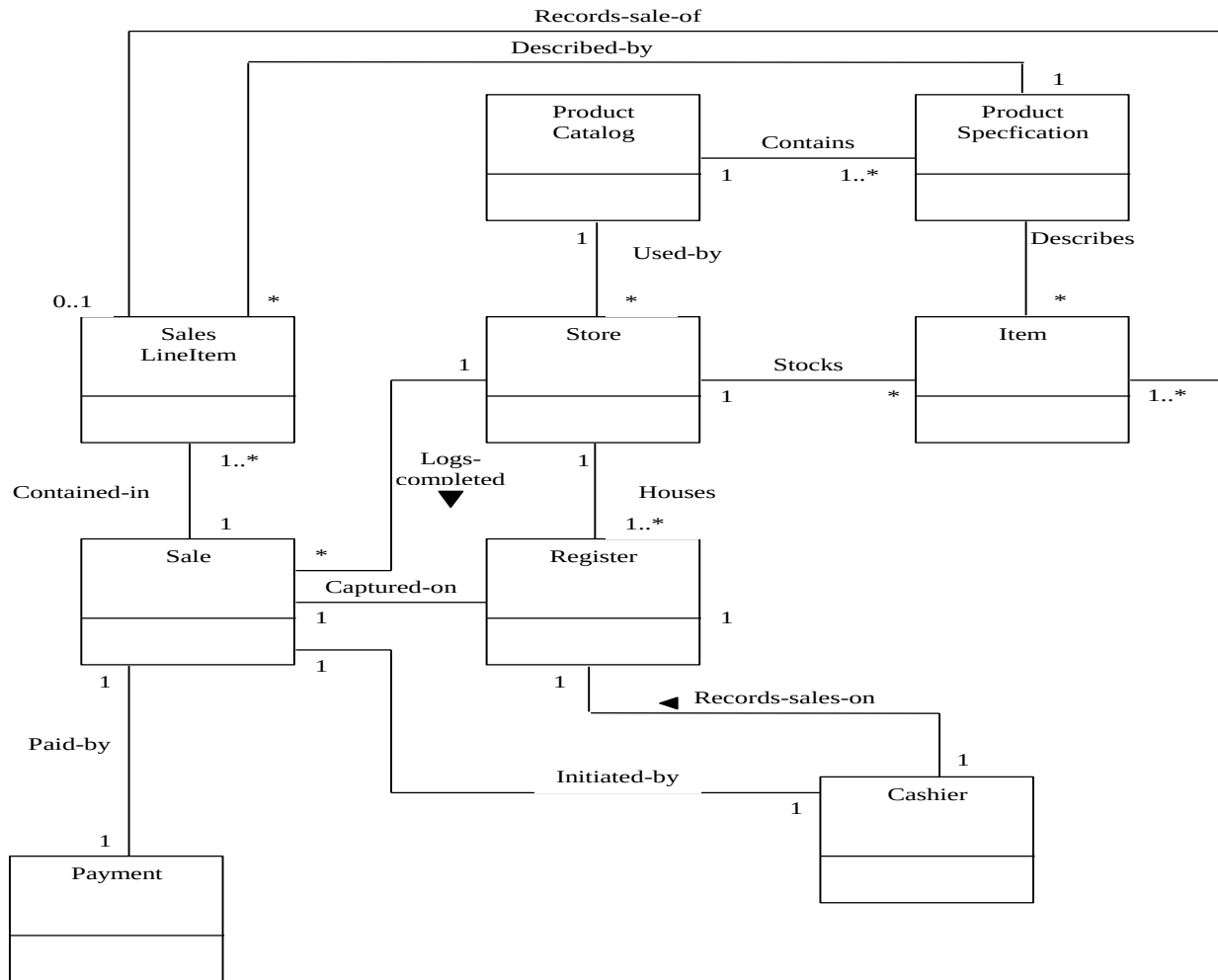
Cleaning Associations 2

- Add comprehension-only associations to enrich critical understanding of the domain.

A partial domain model



Without need-to-know associations



Conclusion

- When in doubt if the concept is required, keep the concept.
- When in doubt if the the association is required, drop it.
- Do not keep derivable association.